

Board-Frame Orientation

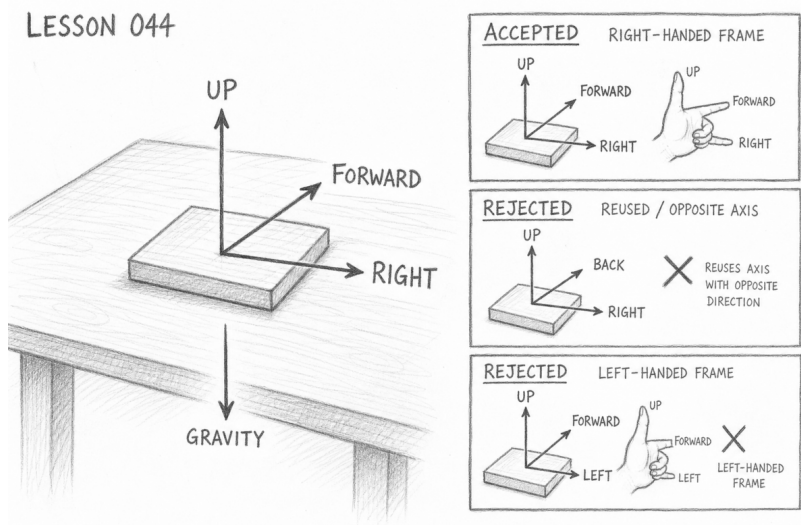
Turn copied gravity into bounded presentation intent

E0 SYNTHETIC REPLAY — NO POWERED INERTIAL MODULE CLAIM

Lesson	044 — component	Time	90–120 minutes
Prerequisite	Lesson 043 copied observations	Board	compile-only Mega replay
Evidence	named memory result cells	Boundary	no bus, pins, wiring, or sensor

Responsibility. Map one current, nonsaturated copied observation into gravity-relative pitch and roll, then map that estimate into light and tone *intent*. The policies own no endpoint or clock. They do not estimate yaw, heading, position, velocity, gestures, or navigation.

1 Name the board frame



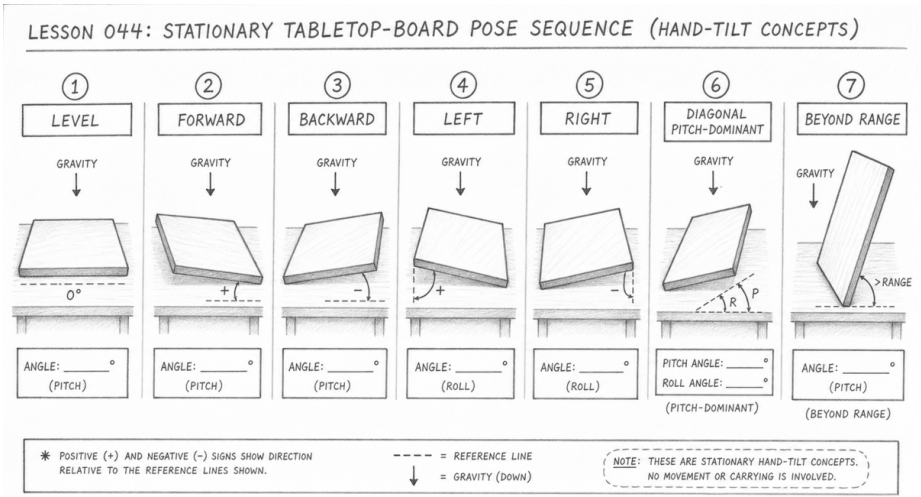
Alternative text: a pencil drawing labels right, forward, and up on a stationary tabletop board, then shows gravity resolved into those three axes for pitch and roll.

Mounting is configuration, not an assumption. Choose three distinct signed sensor axes for board right *R*, forward *F*, and up *U*. They must form a right-handed frame. Duplicate axes, opposite reuse of one axis, and left-handed frames are rejected during initialization.

Board direction	Signed sensor axis	How you checked right-handedness
right	_____	_____
forward	_____	_____
up	_____	_____

Only a **Current** Lesson 043 observation with both readiness fields true, nonzero configuration, calibration, and freshness revisions, nonzero maximum age, no saturation, and OK status is eligible. Gravity magnitude outside its configured band or any rate beyond the stationary threshold yields **Unsteady**, not a motion estimate.

2 Predict poses before calculating



Alternative text: pencil sketches show a level board, four cardinal tilts, and a diagonal tilt, with positive pitch pointing forward and positive roll pointing right.

For remapped acceleration R, F, U , work in micro-g:

$$h = \lfloor \sqrt{R^2 + U^2} \rfloor, \quad pitch = \text{atan2}(F, h), \quad roll = \text{atan2}(R, U).$$

The implementation uses widened integer arithmetic and a normalized 16-step CORDIC. Its test oracle permits at most 4 millidegrees error. That error is included conservatively at level and presentation thresholds.

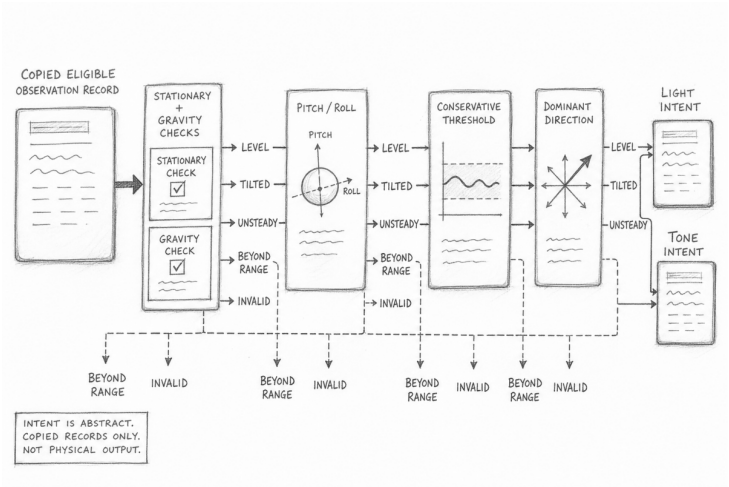
Pose	$R\ (\mu g)$	$F\ (\mu g)$	$U\ (\mu g)$	Prediction
level	_____	_____	_____	_____
forward	_____	_____	_____	_____
right	_____	_____	_____	_____
diagonal	_____	_____	_____	_____

Fixed-point worksheet

$h\ (\mu g)$	pitch (mdeg)	roll (mdeg)	quality
_____	_____	_____	_____
_____	_____	_____	_____

Both absolute angles inside the conservative level bound yield **Level**. An angle in a threshold error band stays **Tilted**; an angle outside the conservative maximum yields **BeyondPresentationRange**. At a diagonal tie, pitch wins: positive pitch is **Forward**; negative pitch is **Backward**. Positive roll is **Right**; negative roll is **Left**.

3 Convert an estimate into intent



Alternative text: a pencil flow maps level, four tilt directions, unsteady, beyond-range, and invalid estimates to complete RGB and tone intent records without drawing or claiming physical endpoints.

Quality	Complete light intent	Tone intent
Level	configured level frame	disabled
Tilted	directional frame, scaled and clamped	only on direction change
Unsteady	phase A or B supplied by caller	disabled
BeyondPresentationRange	configured fault-safe frame	disabled
Invalid/non-OK	configured invalid frame	disabled

For a tilted estimate, calculate the unclamped intensity:

$$I = \frac{|dominantAngle| \times sensitivityPer mille}{fullScaleAngleMilliDegrees}.$$

All angles are millidegrees and sensitivity is 1–1000 permille. The widened product prevents overflow. Clamp *I* to the configured minimum and maximum; do not wrap it.

dominant angle (mdeg)	sensitivity (permille)	full scale (mdeg)	intensity (permille)
_____	_____	_____	_____
_____	_____	_____	_____

A bounded tone is intent, not sound. It is enabled only when a valid tilted direction changes. Replaying the same direction is silent. The caller supplies `diagnosticPhase`; the policy contains no hidden timer and never toggles an LED.

4 Replay and inspect named result cells

LESSON 044: MEMORY-RECORD LEDGER (PENCIL REPLAY)				
LEVEL	PREDICT	ESTIMATE	INTENT	INTERPRET
FORWARD				
BACKWARD				
LEFT				
RIGHT				
DIAGONAL				
UNSTEADY				
BEYOND				
INVALID	SILENT → RESET			

Alternative text: a pencil worksheet sends synthetic level, cardinal, diagonal, unsteady, beyond-range, and invalid frames through orientation and presentation policies into named memory result cells.

Use the Lesson 044 compile-only replay. Predict first, then copy the complete orientation and presentation records. A result cell is host-harness or simulator evidence; it is not a physical light, sound, or sensor observation.

Frame	Orientation prediction	Intent prediction	Observed/interpretation
fixed self-test	_____	_____	_____
level	_____	_____	_____
four cardinals	_____	_____	_____
pitch-dominant diagonal	_____	_____	_____
unsteady phase B	_____	_____	_____
beyond range	_____	_____	_____
invalid	_____	_____	_____

Replay boundary. Mega compilation proves packaging only. Do not run this E0 sketch on a powered board. It owns no sensor bus, RGB pin, mono diagnostic, sounder, resistor, wiring, or electrical test point.

5 Try controlled changes

1. Swap two board axes. Predict initialization rejection.

2. Raise one angular-rate axis one unit above its threshold. Predict **Unsteady** and disabled tone.

3. Replay a tilt with sensitivities 1 and 1000 permille. Calculate both intensities, including clamps.

4. Repeat one direction, then change it. Predict silence, then one bounded direction-change tone intent.

6 Diagnose without inventing motion

Evidence	Diagnosis
stale, saturated, or latest data not ready	invalid orientation; no angles presented
gravity outside band or excessive rate	Unsteady ; stationary assumption failed
angle in maximum-range error band	BeyondPresentationRange
invalid estimate or non-OK status	red/fault intent; tone disabled
MPU6050 or QMI8658 name	future adapter seam, not E0 support

7 Separate acquisition from safe state

Stage	Predict	Observe and interpret
Acquire	initialize valid frame and bounds	status/result cells: _____
Observe	derive from one eligible synthetic frame	estimate/intent: _____
Reset	canonical invalid estimate and fault intent	snapshot: _____

Initialization is resource-acquisition evidence only for pure software objects. Reset is their software safe state. Neither proves bus rollback, electrical shutdown, pin state, or power removal.

8 Blank E1 acceptance record

Do not prefill from compilation, replay, a datasheet, or this PDF.

Required field	Bench record
board revision / exact specimen / markings	_____
supply / logic levels / instrument	_____
tool version / tested commit	_____
resource-acquisition prediction / observation	_____
safe-state prediction / observation	_____
interpretation / reviewer / date	_____

9 Evidence links

- Orientation and presentation public interface
- Deterministic Lesson 044 tests
- Compile-only synthetic replay
- Lessons 043–045 design and open gates

The searchable HTML lesson is the hardware-independent accessible route. This untagged print edition makes no PDF/UA, powered-module, or physical acceptance claim.